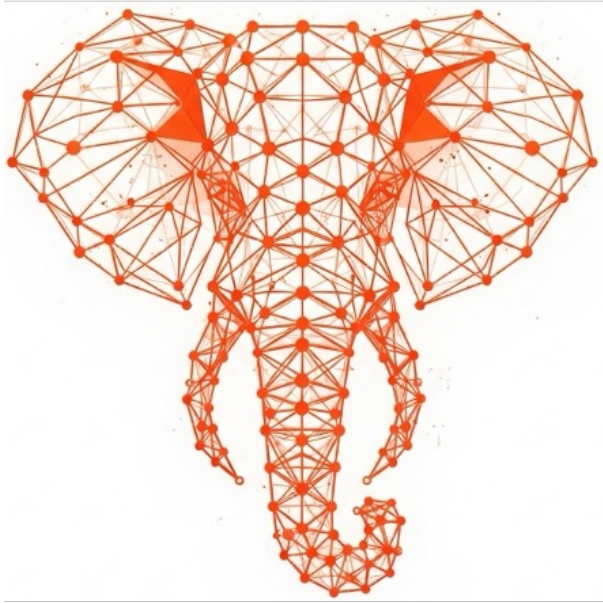




Uni
ct **DEEP LEARNING**
ADVANCED MODELS AND METHODS

Deep Learning

Advanced Models & Methods

Prof. Antonino Furnari (antonino.furnari@unict.it)

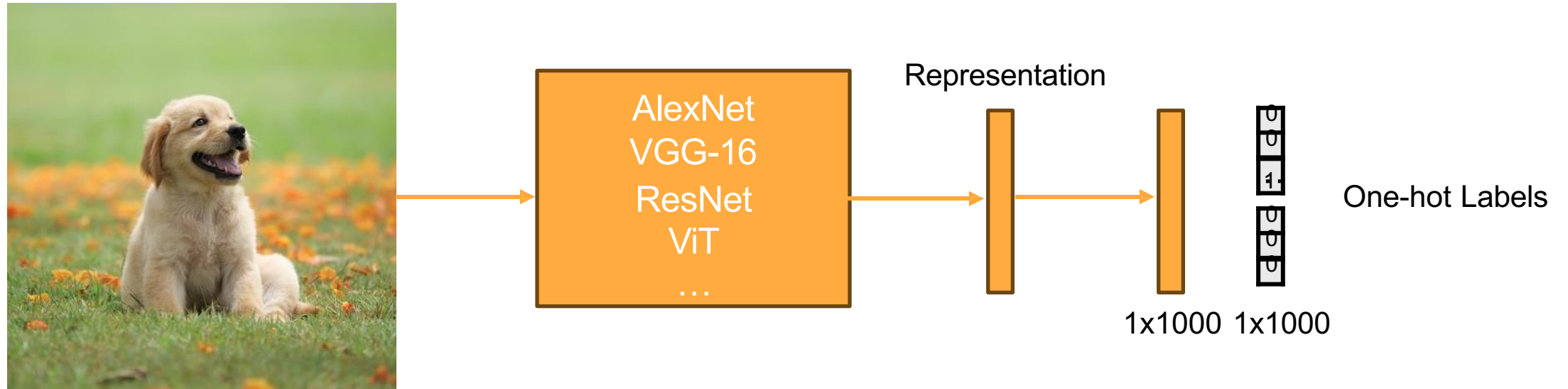
Corso di Laurea Magistrale in Informatica

Dip. di Matematica e Informatica

Università di Catania

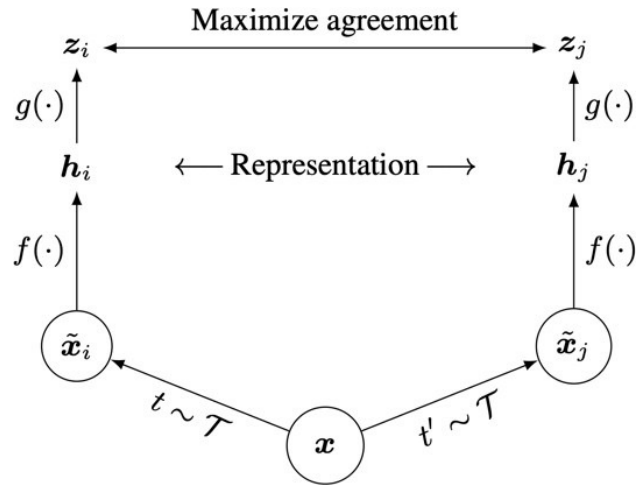
Vision & Language

So Far: Fixed Taxonomies

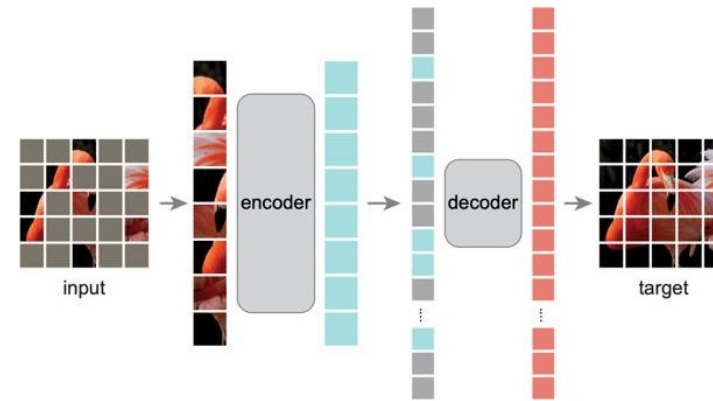


- Models are trained to predict a fixed set of predetermined object categories
- Require crowd-sourced labeling of “gold labels”

Self-Supervised Learning



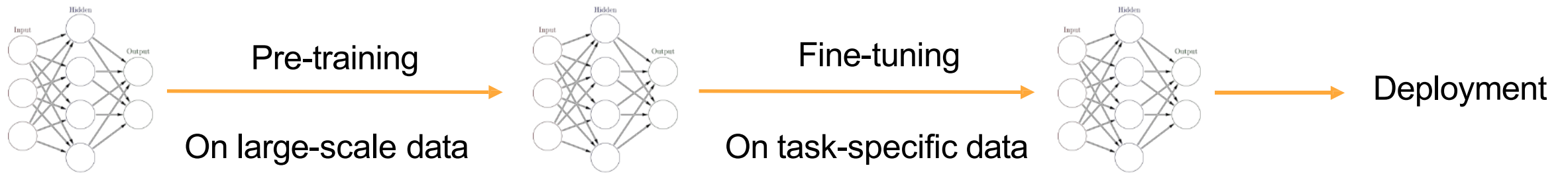
SimCLR (Chen et al. 2020)



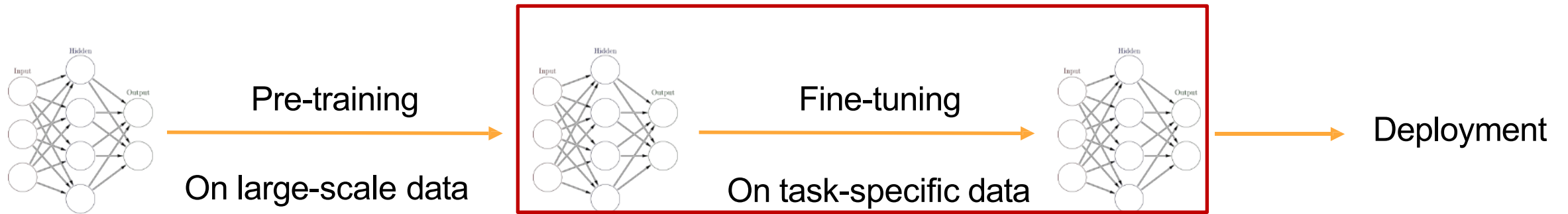
MAE (He et al. 2021)

learning to extract powerful representations without human annotated labels

Adapt to Downstream Task

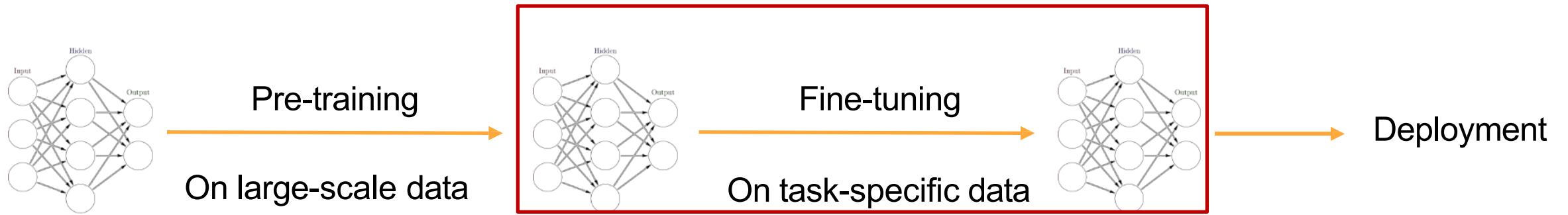


Adapt to Downstream Task



- Require task-specific annotations
- Reduce robustness

Adapt to Downstream Task



How to skip this step to enable zero-shot generalization?

Natural Language Supervision

WIKIPEDIA
The Free Encyclopedia

Photograph a historic site, help Wikipedia, and win a prize. Participate in the world's largest photography competition this month!
[Learn more](#)

Golden Retriever

63 languages

Article

Talk

Read

View source

View history

Tools

Contents

hide

(Top)

History

Description

Appearance

Temperament

Use

Health

Notable Golden Retrievers

Explanatory notes

References

Further reading

From Wikipedia, the free encyclopedia

For other uses, see [Golden Retriever \(disambiguation\)](#).

The **Golden Retriever** is a Scottish [breed](#) of [retriever](#) dog of medium size. It is characterised by a gentle and affectionate nature and a striking golden coat. It is a working dog, and registration is subject to successful completion of a [working trial](#).^[2] It is commonly kept as a [pet](#) and is among the most frequently registered breeds in several Western countries; some may compete in [dog shows](#) or [obedience trials](#), or work as a [guide dog](#).

The Golden Retriever was bred by [Sir Dudley Marjoribanks](#) at his Scottish estate [Guisachan](#) in the late nineteenth century. He [cross-bred](#) [Flat-coated Retrievers](#) with [Tweed Water Spaniels](#), with some further infusions of [Red Setter](#), [Labrador Retriever](#) and [Bloodhound](#). It was recognised by the [Kennel Club](#) in 1913, and during the [interwar period](#) spread to many parts of the world.

Golden Retriever

Other names

Flat-coated Retriever, Golden, Yellow or Golden Retriever

Origin

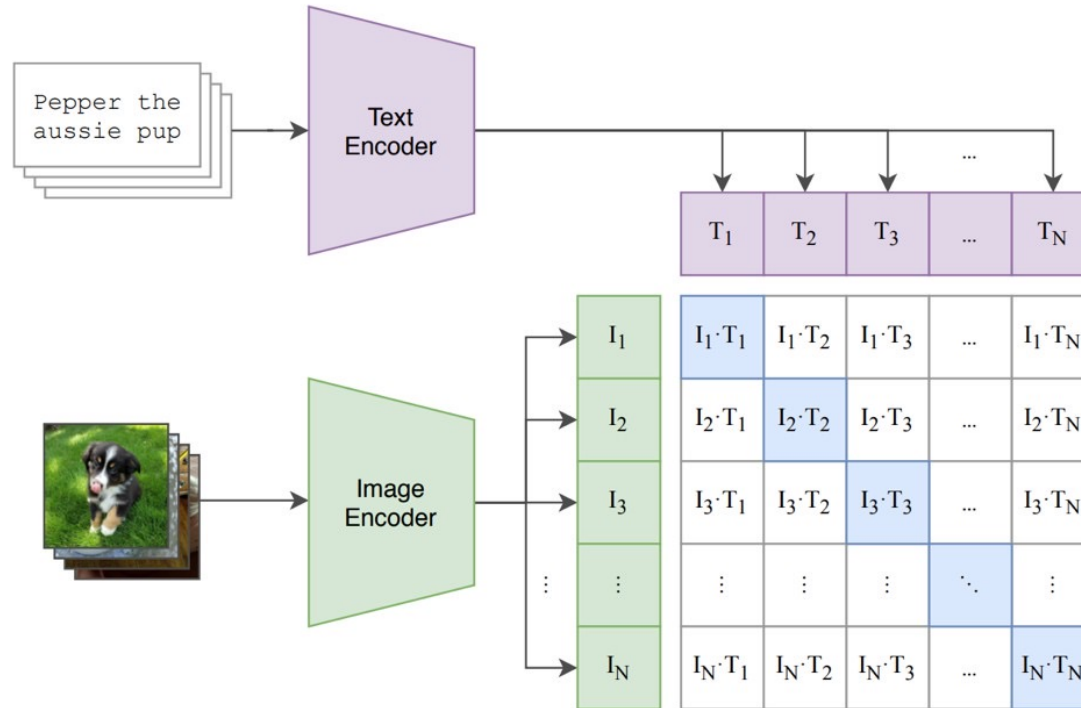
Scotland

➤ Unlimited Data: vast amount of text-image pairs on the internet

➤ Flexible zero-shot transfer: learn a representation that connects visual content to language

Overview of CLIP

(1) Contrastive pre-training

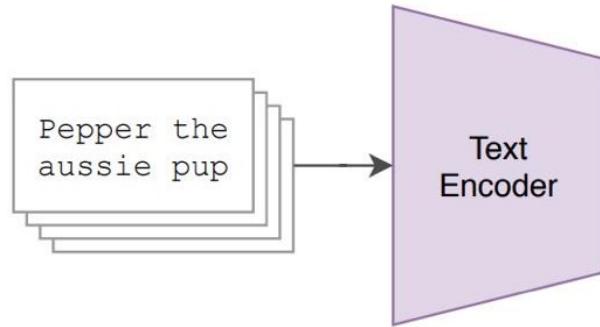


- Data: 400 million image-text pairs collected from the internet
- Model: ResNets and ViTs with up to 300M parameters
- Loss Function: Contrastive Loss
- Compute: Trained on 250 - 600 GPUs for up to 18 days.

CLIP: Contrastive Language-Image Pre-training

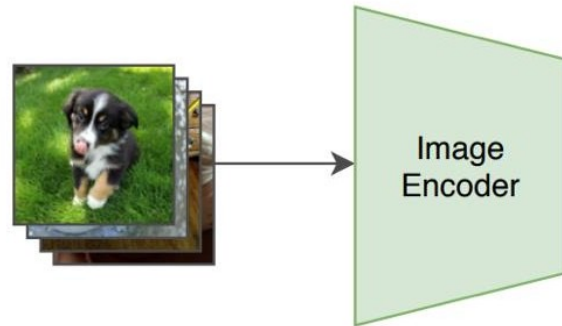
CLIP (Radford et al. 2021)

Encoders for CLIP



- Transformer
- 12- layer 512/768 wide model with 8/12 attention heads
- Max sequence length was capped at 76

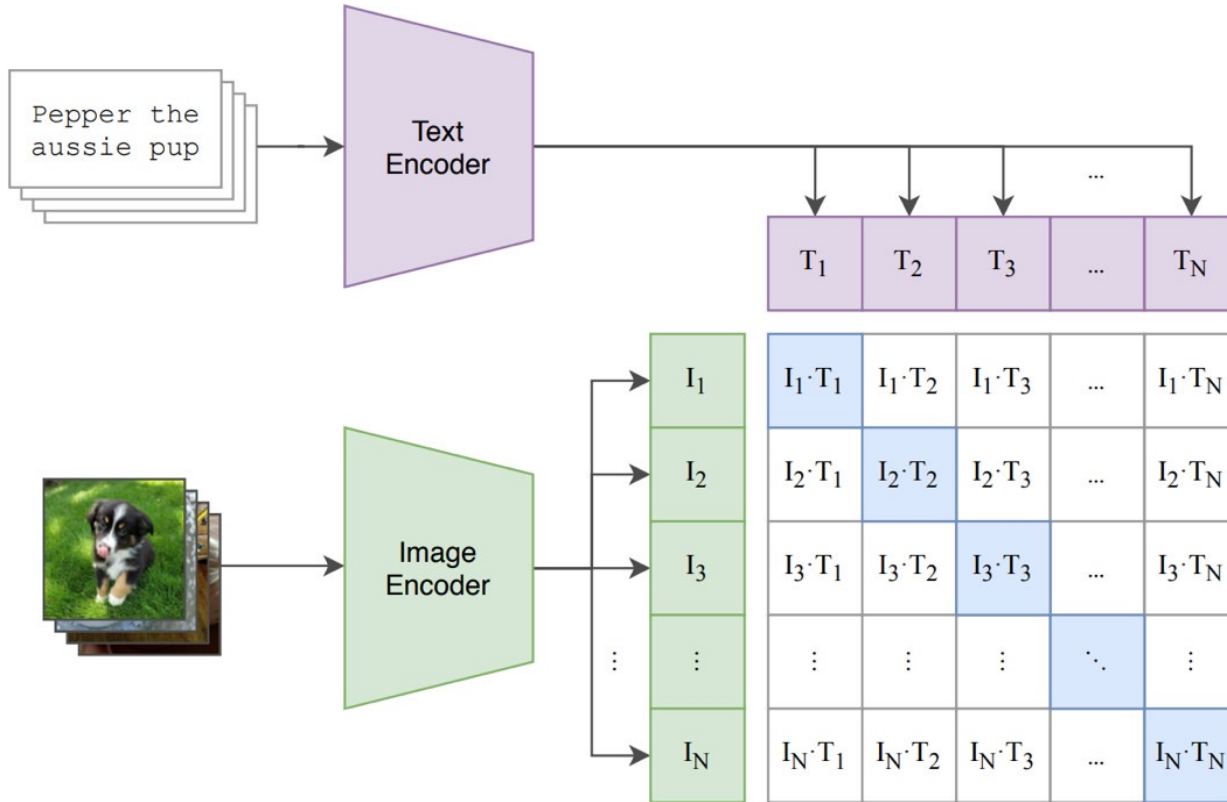
- **Weights are trained from scratch**



- ResNet-50, ResNet-101, ViT-B/32, ViT-B/16, ViT-L/14
- Pre-train at a higher resolution for one additional epoch ViT-L/14@336px (**Best Model**)
- 307M parameters

CLIP (Radford et al. 2021)

Contrastive Loss for CLIP



Positive Pairs:

Aligned text-images in a minibatch $N \times 1$

Negative Pairs:

Mis-aligned text-images in a minibatch $N \times N - N$

CLIP (Radford et al. 2021)

Contrastive Loss for CLIP



A photo of a golden retriever

x_1

y_1



A lake surrounded by mountains
and woods

x_2

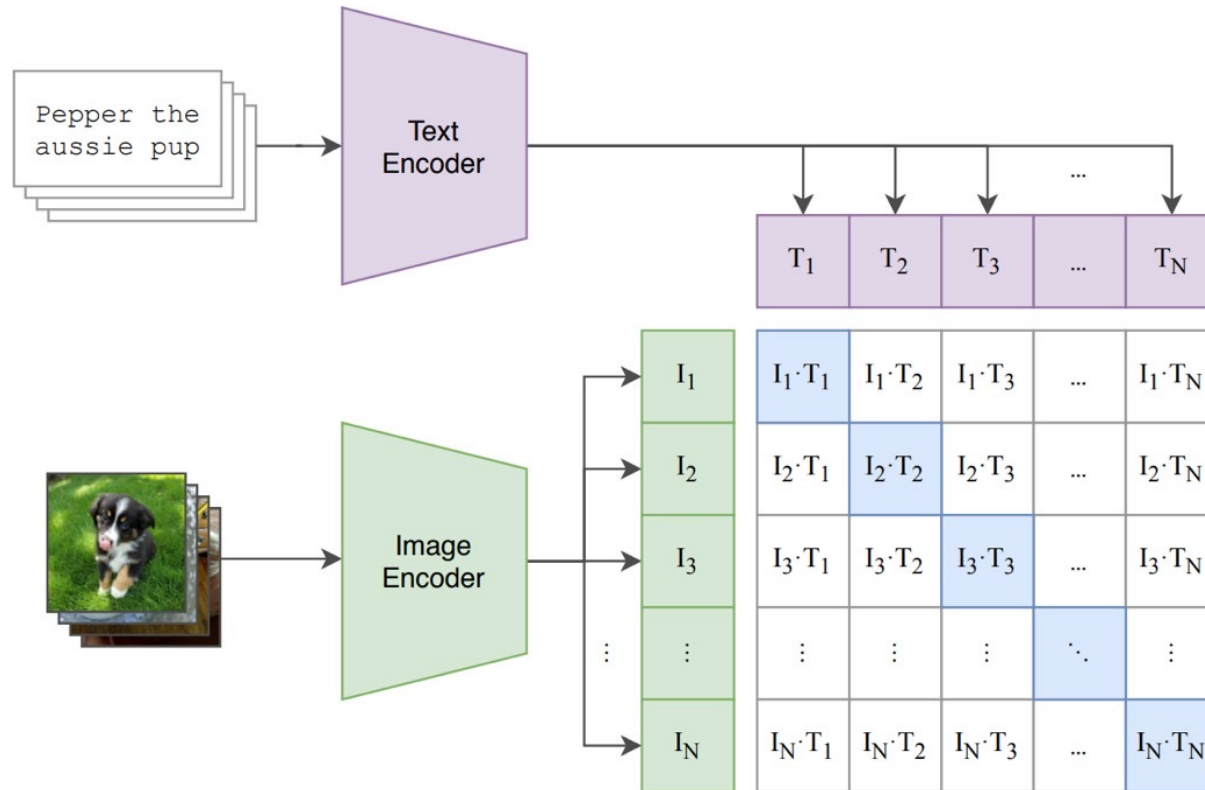
y_2

CLIP (Radford et al. 2021)

$$L_{i2t} = -\frac{1}{N} \sum_i \log \frac{\exp(x_i^\top y_i / \tau)}{\sum_{j=1}^N \exp(x_i^\top y_j / \tau)}$$

$$L_{t2i} = -\frac{1}{N} \sum_i \log \frac{\exp(y_i^\top x_i / \tau)}{\sum_{j=1}^N \exp(y_i^\top x_j / \tau)}$$

Contrastive Loss for CLIP



```
# image_encoder - ResNet or Vision Transformer
# text_encoder - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l] - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T) #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss = (loss_i + loss_t)/2
```

CLIP (Radford et al. 2021)

Data Curation for CLIP

WebImageText (WIT) OpenAI Private Dataset

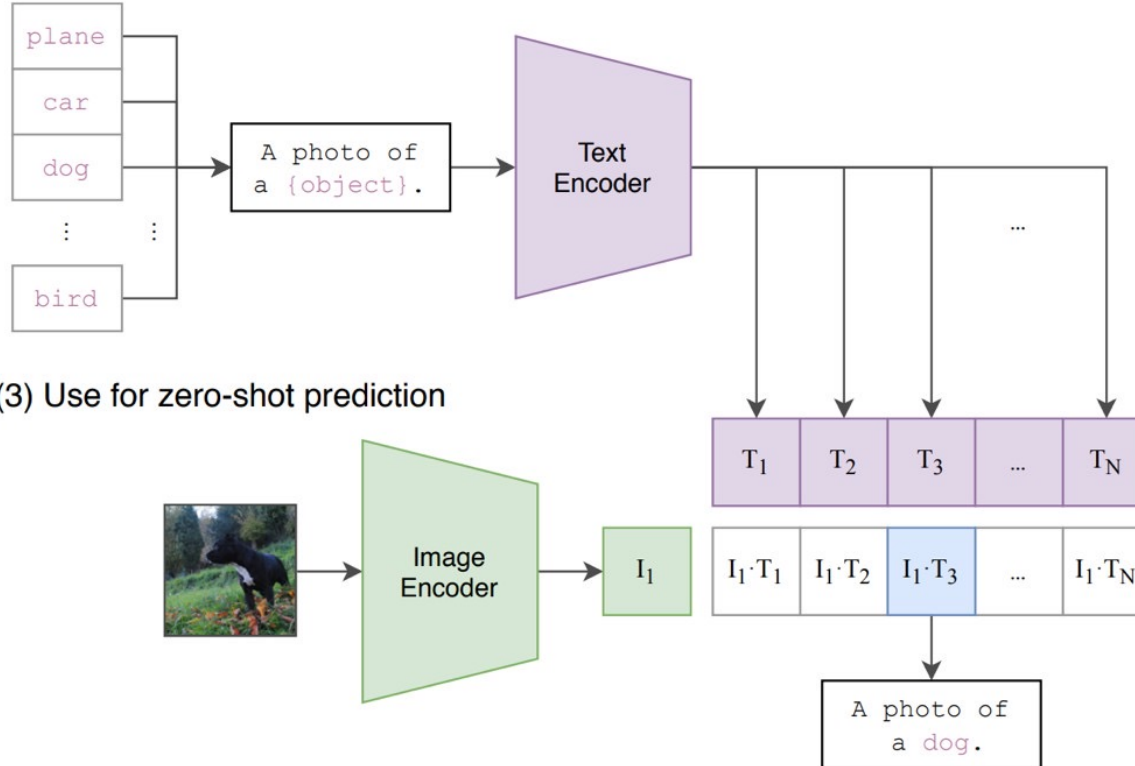
- 400M image-text data pairs
- search for (image, text) pairs as part of the construction process whose text includes one of a set of 500,000 queries
- balance the results by including up to 20,000 (image, text) pairs per query

[Demystifying CLIP Data](#) (Xu et al. ICLR 2024)

CLIP (Radford et al. 2021)

Zero-shot Prediction

(2) Create dataset classifier from label text



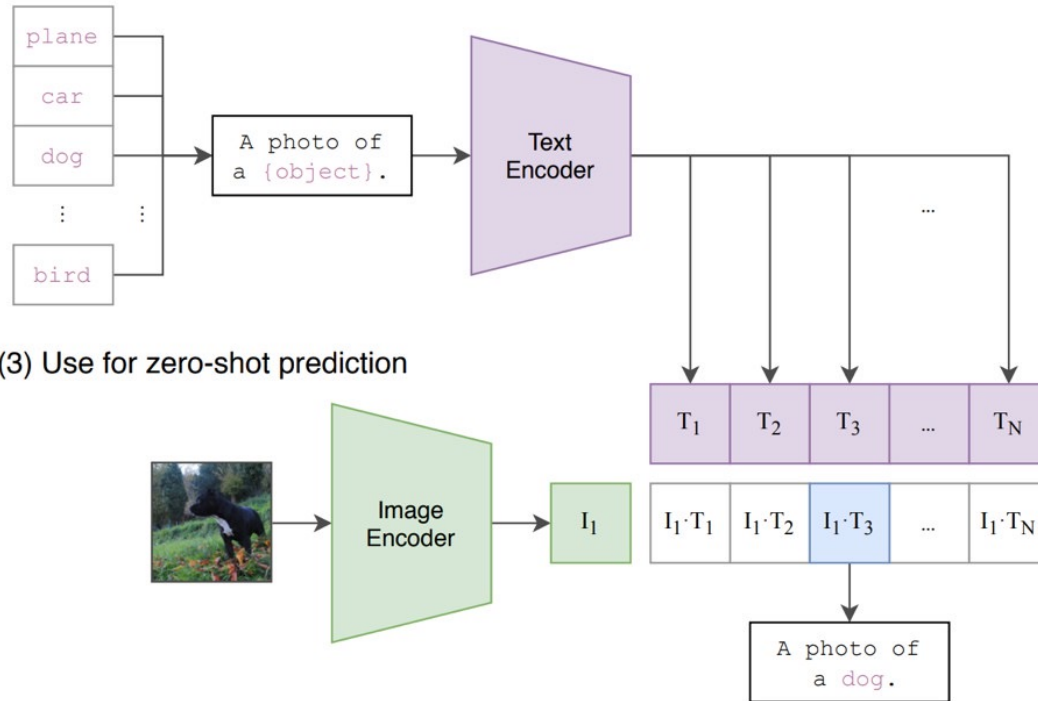
(3) Use for zero-shot prediction

- Construct a prompt template “A photo of a {label}” rather than “{label}” (1.3% ↑)
 - Text with a single word relatively rare in the pre-training dataset
- Provide domain-specific context “A photo of a {label}, a type of pet.” for Oxford Pets
- Prompt ensembling. 80 different context prompts ensemble for ImageNet (3.5% ↑)

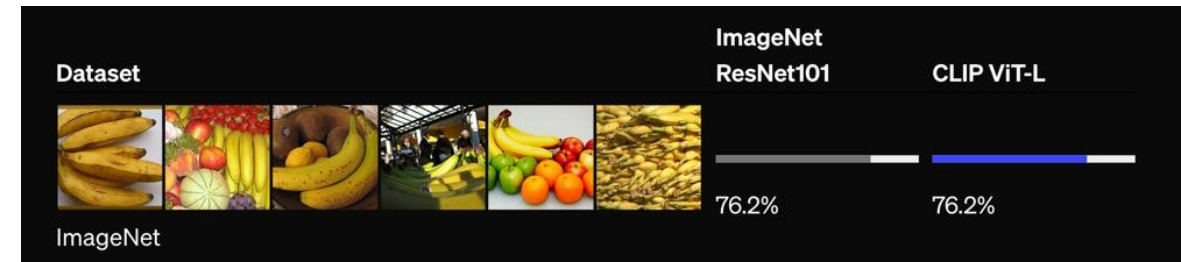
CLIP (Radford et al. 2021)

Zero-shot Prediction

(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

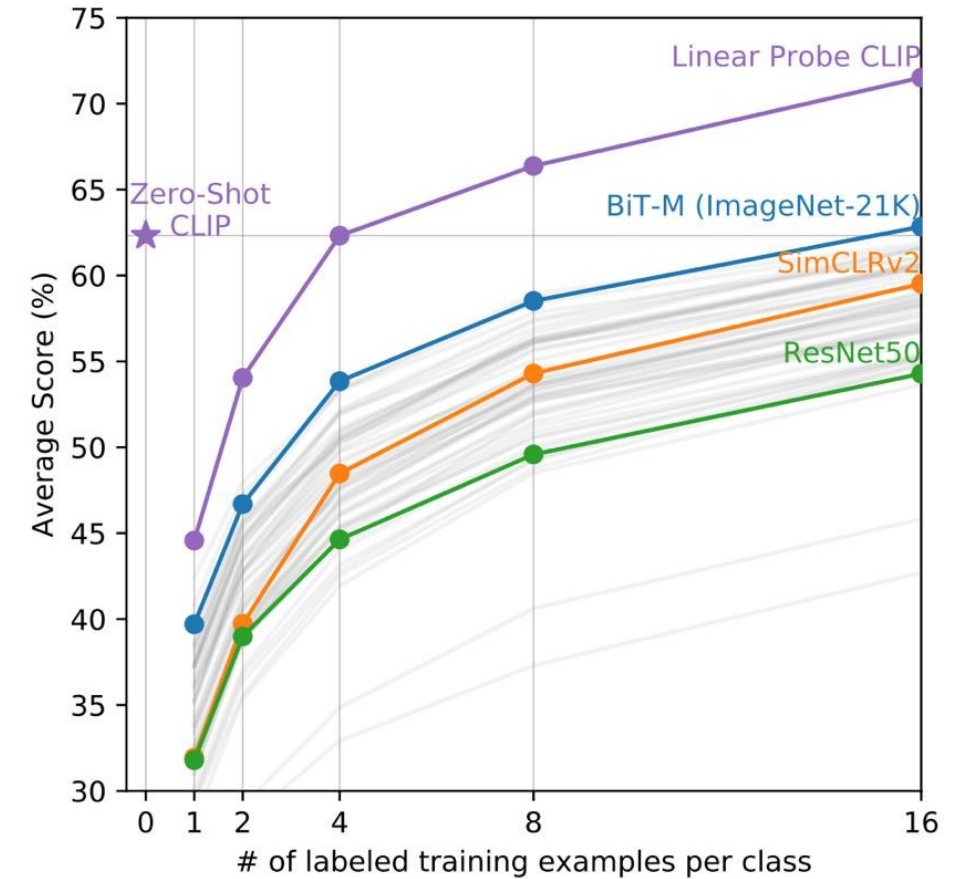


Comparable performance on ImageNet without training on that!

CLIP (Radford et al. 2021)

Robustness to Distribution Shift and Zero-Shot Probing

	Dataset Examples	ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet		76.2	76.2	0%
ImageNetV2		64.3	70.1	+5.8%
ImageNet-R		37.7	88.9	+51.2%
ObjectNet		32.6	72.3	+39.7%
ImageNet Sketch		25.2	60.2	+35.0%
ImageNet-A		2.7	77.1	+74.4%



CLIP (Radford et al. 2021)

Video & Language

Sources of Video & Language Data

Video and language data can be found in different forms:



Instructional videos from YouTube with transcribed speech

Egocentric video with dense manual textual descriptions



C separates the yarn.



C lifts container.

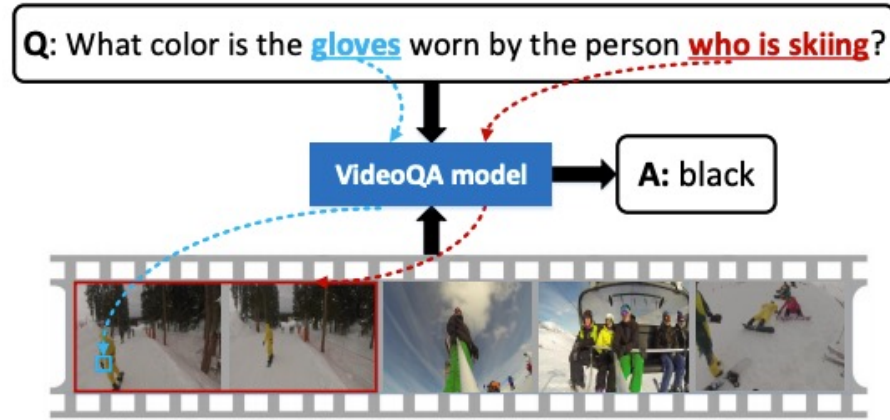


C operates the camera.

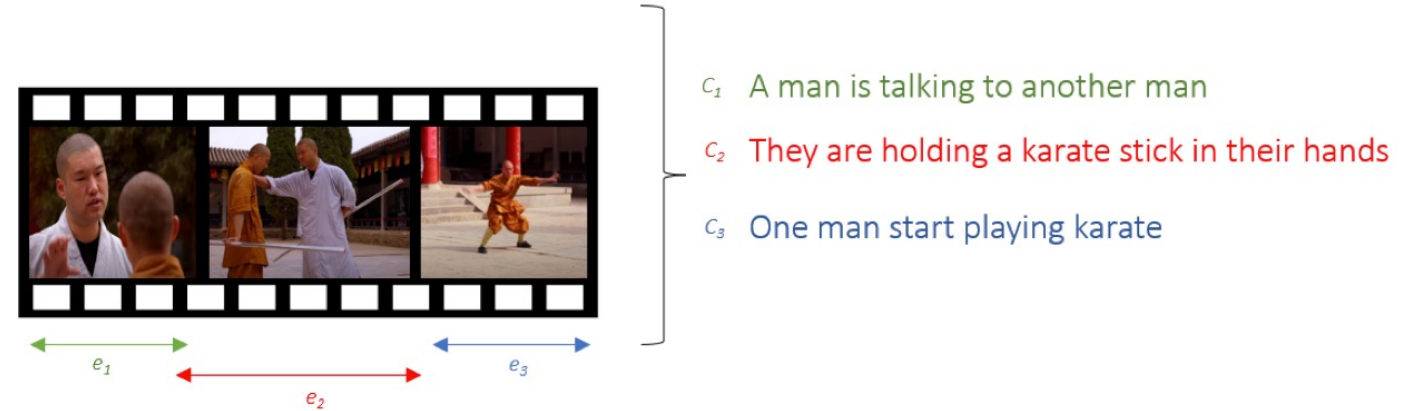


Movies with subtitles

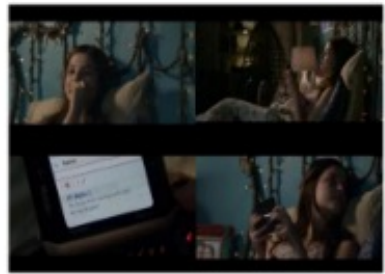
Video & Language Tasks



Video Question-Answering

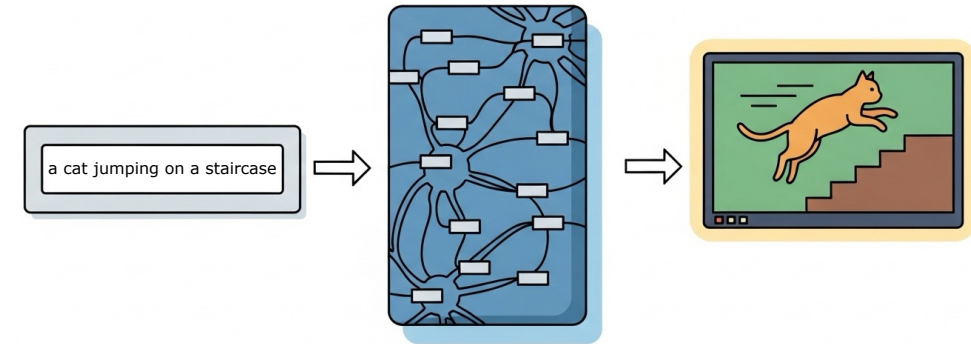


Video Captioning



1. A woman is laying in bed.
2. A woman is talking about her hair.
3. Someone is talking in bed.

Video-Text Retrieval



Video-Text Generation

N. Madan, A. Moegelmose, R. Modi, Y. S. Rawat, e T. B. Moeslund, «Foundation Models for Video Understanding: A Survey», [10.48550/arXiv.2405.03770](https://arxiv.org/abs/2405.03770).

I. Qasim, A. Horsch, e D. K. Prasad, «Dense Video Captioning: A Survey of Techniques, Datasets and Evaluation Protocols», 2023, *arXiv*: arXiv:2311.02538

Yu, Zhou, et al. "Activitynet-qa: A dataset for understanding complex web videos via question answering." *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 33. No. 01. 2019.

Learning Aligned Video-Text Representations



HowTo100M is a large-scale dataset of narrated videos with an emphasis on instructional videos where content creators teach complex tasks with an explicit intention of explaining the visual content on screen. HowTo100M features a total of:

- **136M video clips with captions** sourced from 1.2M Youtube videos (15 years of video)
- **23k activities** from domains such as cooking, hand crafting, personal care, gardening or fitness

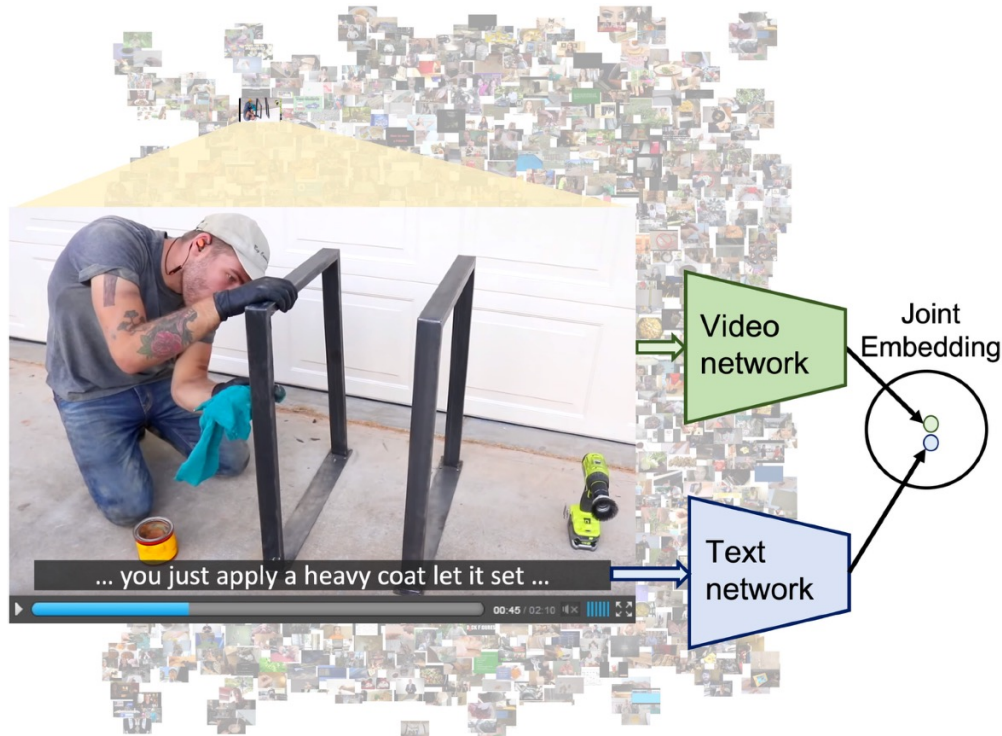
Each video is associated with a narration available as subtitles automatically downloaded from Youtube.

- Videos were sourced from YouTube using wikiHow article titles as search keyword for 23K how-to tasks from 12 domains that are particularly visual

Learning a joint instructional video-text embedding



- A simple contrastive training recipe:
 - large similarity for **positive** (modality₁, modality₂) pairs
 - small similarity for **negative** (modality₁, modality₂) pairs



Such space enables:

- Video-text retrieval
- Open world action classification (see what's the closest action label)
- Language-aware representation learning (can fine-tune the model to a new task later)
- Video-text alignment (e.g., for subtitles)

Learning a joint instructional video-text embedding



- A simple contrastive training recipe:
 - large similarity for **positive** (modality₁, modality₂) pairs
 - small similarity for **negative** (modality₁, modality₂) pairs

Max-margin ranking loss

$$\sum_{i \in \mathcal{B}} \sum_{j \in \mathcal{N}(i)} \max(0, \delta + s_{i,j} - s_{i,i}) + \max(0, \delta + s_{j,i} - s_{i,i}),$$

where $s_{i,j} = s(V_i, C_j)$ is the similarity score $s(V, C) = \frac{\langle f(\mathbf{v}), g(\mathbf{c}) \rangle}{\|f(\mathbf{v})\|_2 \|g(\mathbf{c})\|_2}$

- Negative samples $\mathcal{N}(i)$
 - 50% selected from the same video
 - 50% chosen from other videos

$f(\mathbf{v})$: two-layer MLP on top of
Kinetics-trained 3D CNN features
 $g(\mathbf{c})$: two-layer MLP on top of
word2vec features

negatives sampled as distant captions/video pairs

Experimental evaluation on video-language tasks



- Qualitative results of text2video retrieval on HowTo100M

Q Iron cloth



Q Cut paper



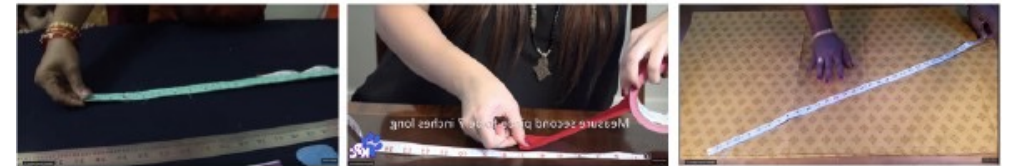
Q Cut wood



Q Crease origami



Q Measure the length



Q Measure blood pressure



Handling temporal misalignment between video and narration



People may talk about a step **before** or **after** they visually demonstrate it!



Some stats collected from HowTo100M [Han et al., CVPR 2022]:

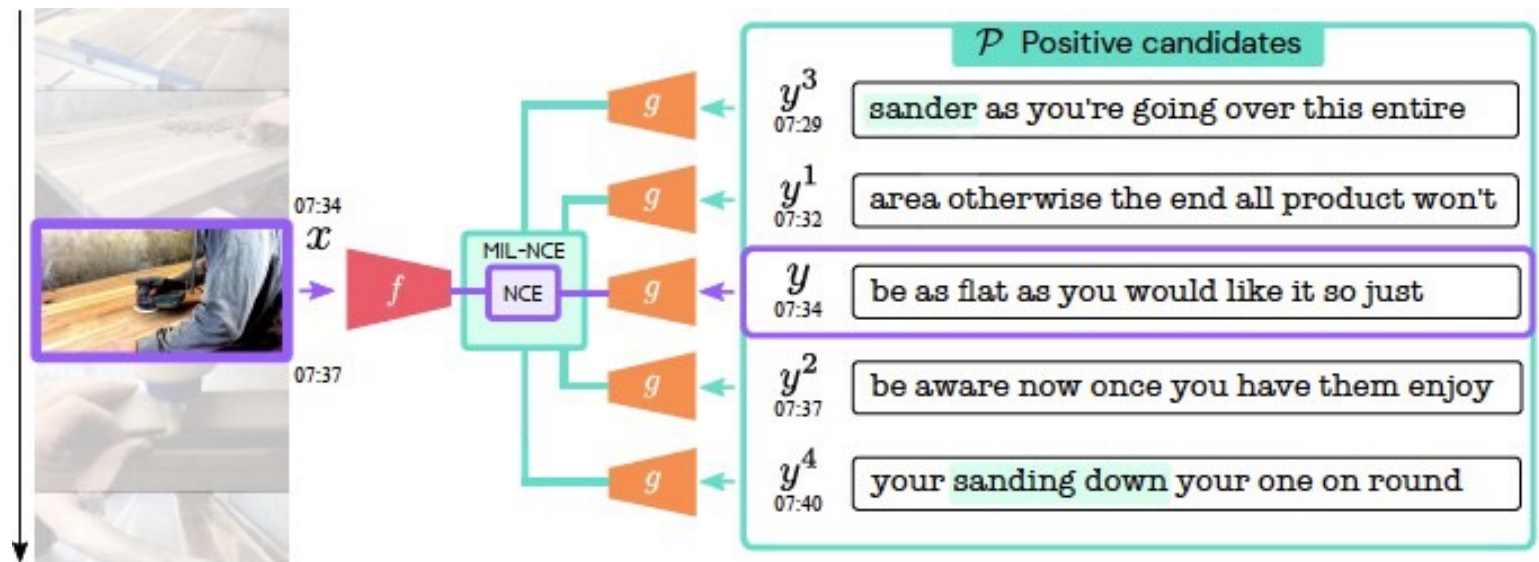
- about 15% of narrations describe an action demonstrated in the same interval
- about 30% of narrations are alignable

Handling temporal misalignment between video and narration



MIL-NCE: Multiple Instance Learning Noise Contrastive Estimation

[Miech et al., CVPR 2020]



**only one match among different candidates
(neighboring captions)**

MIL-NCE: Multiple Instance Learning Noise Contrastive Estimation

[Miech et al., CVPR 2020]

- Joint probability under temporal alignment assumption: $p(x, y; f, g) \propto e^{f(x)^\top g(y)}$
- Relaxing the assumption to temporal adjacency: $p(\cup_k \{(x, y_k)\}) = \sum_k p(x, y_k) \propto \sum_k e^{f(x)^\top g(y_k)}$
- Maximizing ratio between **sum** of positive candidate scores and sum of negative scores:

MIL-NCE

$$\max_{f, g} \sum_{i=1}^n \log \left(\frac{\sum_{(x, y) \in \mathcal{P}_i} e^{f(x)^\top g(y)}}{\sum_{(x, y) \in \mathcal{P}_i} e^{f(x)^\top g(y)} + \sum_{(x', y') \sim \mathcal{N}_i} e^{f(x')^\top g(y')}} \right)$$

accounts for the «multiple instance» assumption – maximize the sum across candidates rather than a single candidate

Note that this differs from traditional NCE, which would assume temporal alignment

$$\max_{f, g} \sum_{i=1}^n \log \left(\frac{e^{f(x_i)^\top g(y_i)}}{e^{f(x_i)^\top g(y_i)} + \sum_{(x', y') \sim \mathcal{N}_i} e^{f(x')^\top g(y')}} \right)$$

Handling temporal misalignment between video and narration

Ablation over training loss [Miech et al., CVPR 2020]

NCE is better than cross-entropy or max-margin:

Loss	YR10	MR10	CTR	HMDB	UCF
Binary-Classif	18.5	23.1	32.6	44.2	68.5
Max margin	16.3	24.1	29.3	56.2	76.6
NCE	29.1	27.0	35.6	55.4	77.5

But MIL-NCE is better than NCE:

$\ \mathcal{P}\ \rightarrow$	NCE	MIL-NCE				
	1	3	5	9	17	33
YR10	29.1	33.6	35.0	33.1	32.4	28.3
MR10	27.0	30.2	31.8	30.5	29.2	30.4
CTR	35.6	37.3	34.2	31.8	25.0	25.0
HMDB	55.4	57.8	56.7	55.7	54.8	51.4
UCF	77.5	79.7	80.4	79.5	78.5	77.9

neighborhood size (for multiple instances)

Beyond Instructional Video: Egocentric Video



Static Third Person Video



Dynamic First Person Video



Characteristics

- 32 kitchens - 4 cities
- Head-mounted camera
- 55 hours of recording - Full HD, 60fps
- 11.5M frames
- Multi-language narrations
- 39,594 action segments
- 454,255 object bounding boxes
- 125 verb classes, 331 noun classes

Suited for capturing long term activities

No scene cuts, content always relevant



Ego4D & Ego-Exo4D



EGO-EXO4D



Ego4D: Around the World in 3,000 hours of Egocentric Video



A Massive-scale, egocentric dataset and benchmark suite.

3,670 hours of daily-life videos

136 indoor + outdoor scenarios

931 unique camera wearers

3.85M human narrations

74 locations across **9** countries

17 tasks across **5** benchmarks

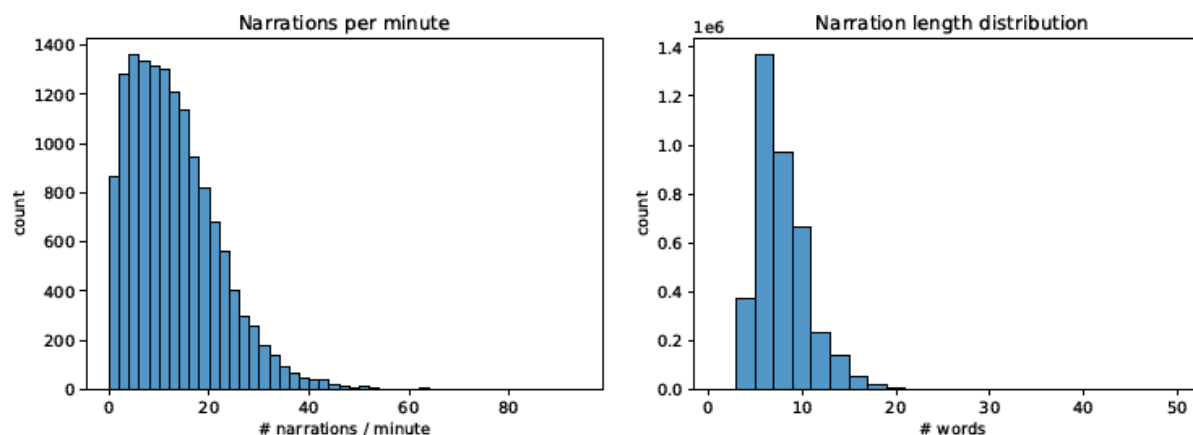


Slide Credit: Lorenzo Torresani

Ego4D narrations



- Designed to provide a play-by-play of the action. Narrator prompt:
“Pretend as you watch this video that you are also talking to a friend on the phone, and you need to describe to your friend everything that is happening in the video. Your friend cannot see the video.”
- Large-scale: 3.85M sentences
- Temporally dense: 13.2 sentences per minute



- Typed (no ASR error!) and manually grounded (no time misalignment!)

Ego4D narrations

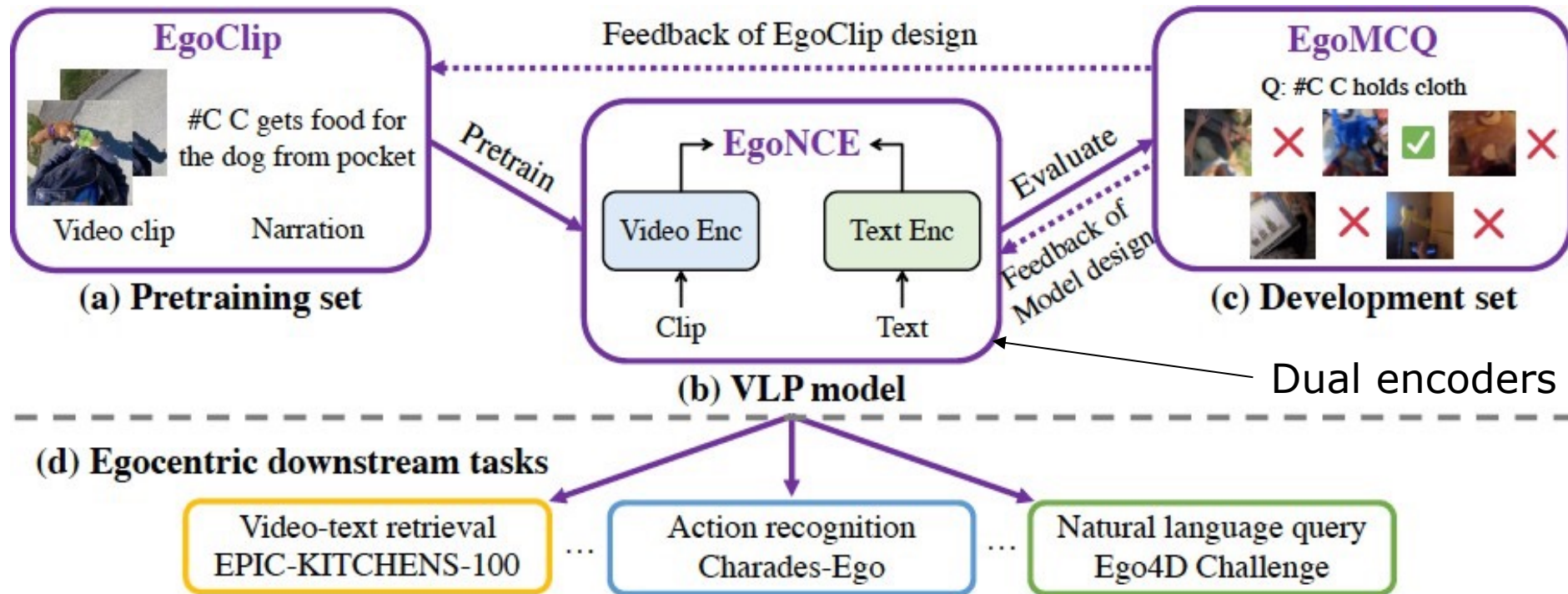


EgoVLP: Egocentric Video-Language Pretraining [Lin et al., NeurIPS 2022]



EgoClip Dataset (sampled from Ego4D)

Development set informs model design



Significant gains on 5 egocentric benchmarks across 3 datasets over existing work:

- [Ego4D Challenges] Object State Change Classification: +5.2% (1st Place)
- [EPIC-KITCHENS Challenges] Multi-Instance Retrieval: +5.9% in nDCG (avg) (1st Place)
- [Ego4D Challenges] Natural Language Query: +5.4% in R@1 (IoU=0.3)
- [Ego4D Challenges] Moment Query: +7.0% in R@1 (IoU=0.3)
- [Charades-Ego] Action Recognition: +2.0% in MAP

EgoNCE loss

Training loss: modified NCE to account for two challenges in egocentric video

1. The **same action** often occurs in **different scenarios**, e.g.,



watching the phone
when **lying on bed**



watching the phone
when **walking outdoors**



positive sampling based on action = <verb + noun>

$$\mathcal{P}_i = \{j \in \mathcal{B} \mid \text{noun}(j) \cap \text{noun}(i) \neq \emptyset, \text{verb}(j) \cap \text{verb}(i) \neq \emptyset\}$$

rather than perfect alignment,
need to share at least one verb
and at least one noun

2. **Different actions** often appear within **same scenario**, e.g.,



moving the mouse
when **working at desk**



typing on the keyboard
when **working at desk**



negative sampling based on temporal adjacency

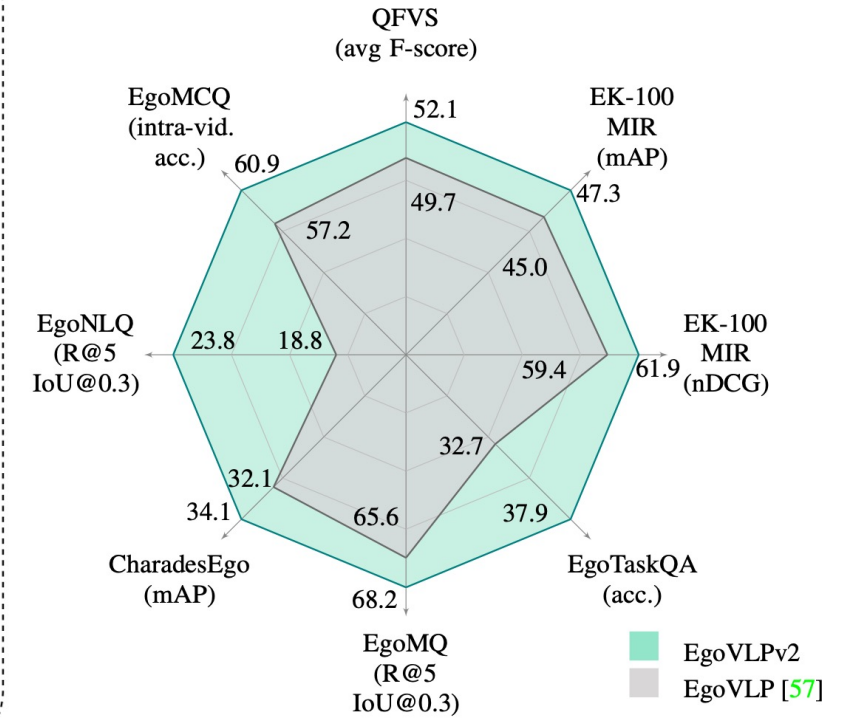
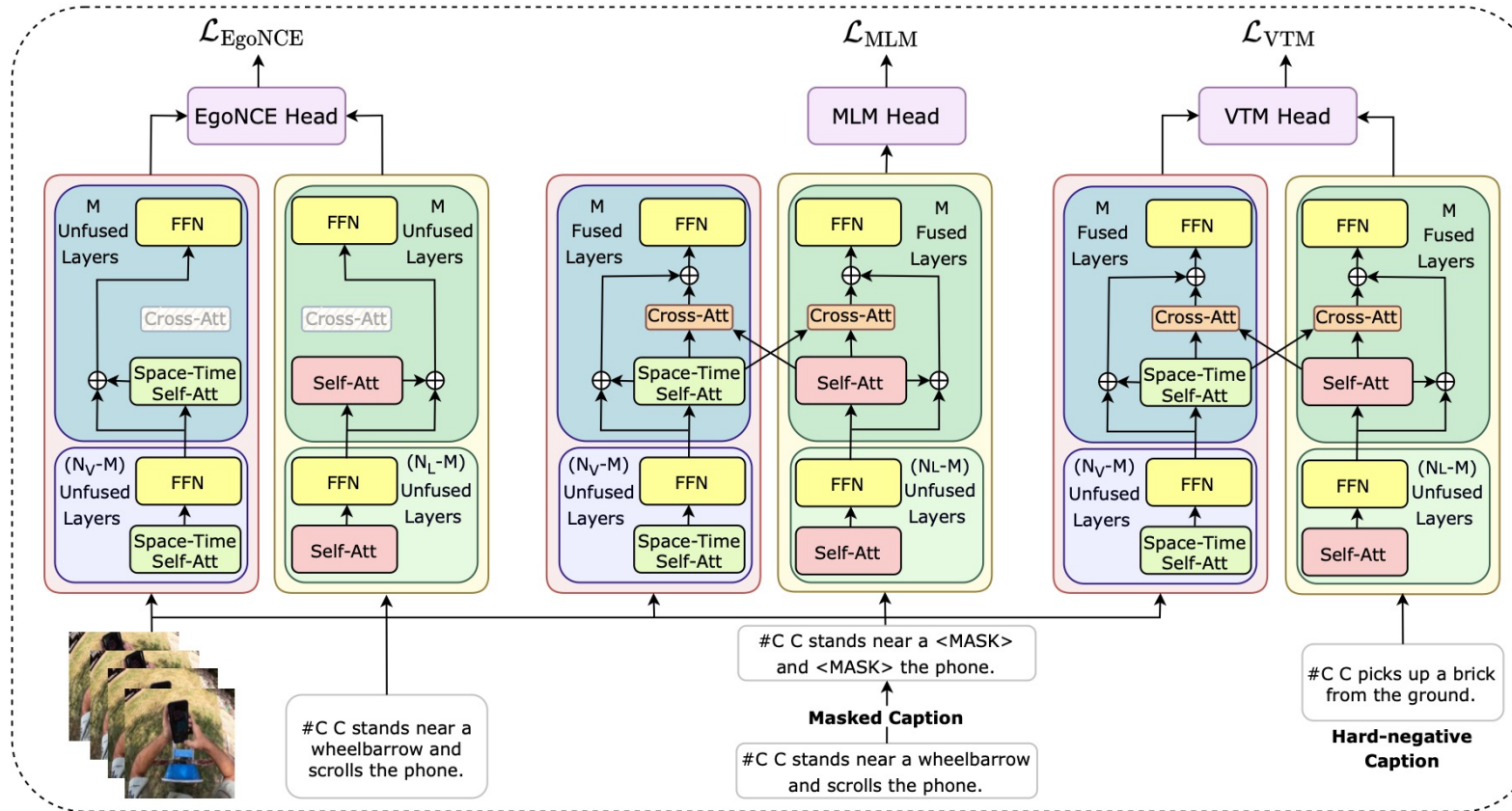
$i' \in \mathcal{N}(i)$: adjacent to i but different action

$$\mathcal{L}_{\text{v2t}}^{\text{ego}} = \frac{1}{|\tilde{\mathcal{B}}|} \sum_{i \in \tilde{\mathcal{B}}} \log \frac{\sum_{k \in \mathcal{P}_i} \exp(\mathbf{v}_i^T \mathbf{t}_k / \tau)}{\sum_{j \in \mathcal{B}} (\exp(\mathbf{v}_i^T \mathbf{t}_j / \tau) + \exp(\mathbf{v}_i^T \mathbf{t}_{j'} / \tau))}.$$

EgoVLPv2

Extends EgoVLP as follows:

- Fusion in the backbone
- Masked caption training
- Hard negatives



Beyond Image & Video

What is ImageBind?

Slide credit: Manish Gupta

- An image of a beach can remind us of the sound of waves, the texture of the sand, a breeze, or even inspire a poem.
- Aligns six modalities' embedding into a common space: images, text, audio, depth, thermal, and Inertial Measurement Unit (IMU).
- Image-paired data is sufficient to bind the modalities together.

1) Cross-Modal Retrieval

Audio	Images & Videos	Depth	Text
 Crackle of a Fire			"A fire crackles while a pan of food is frying on the fire." "Fire is crackling then wind starts blowing." "Firewood crackles then music..."
 Baby Cooing			"A baby is crying while a toddler is laughing." "A baby is laughing while an adult is laughing." "A baby laughs and something..."

2) Embedding-Space Arithmetic



Diagram illustrating Embedding-Space Arithmetic: An image of a white egret is added to the audio embedding of "Waves" (represented by a speaker icon), resulting in a new image of a white egret in water.

3) Audio to Image Generation



Diagram illustrating Audio to Image Generation: Audio inputs (Dog, Engine, Fire, Rain) are converted into corresponding visual images (a dog, a bus, a fire, and rain).

How is the ImageBind model trained?



Slide credit: Manish Gupta

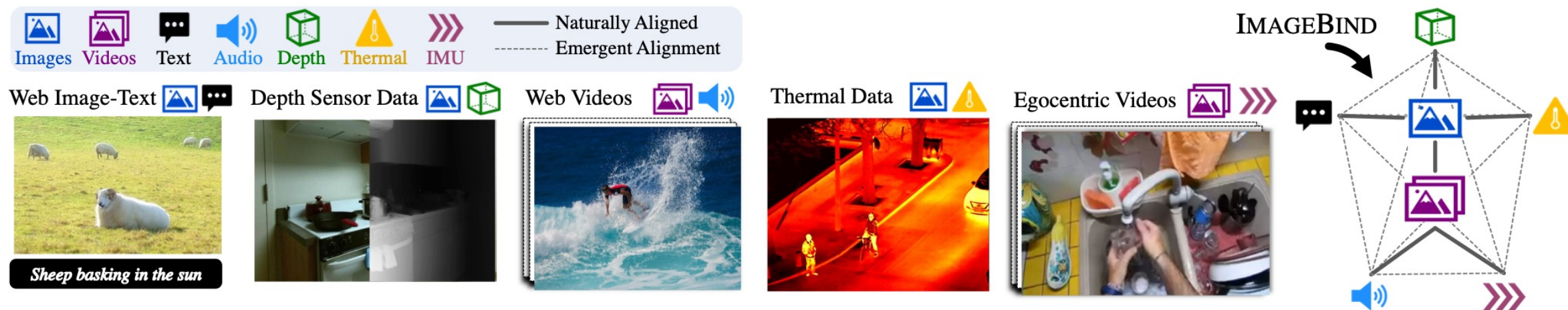
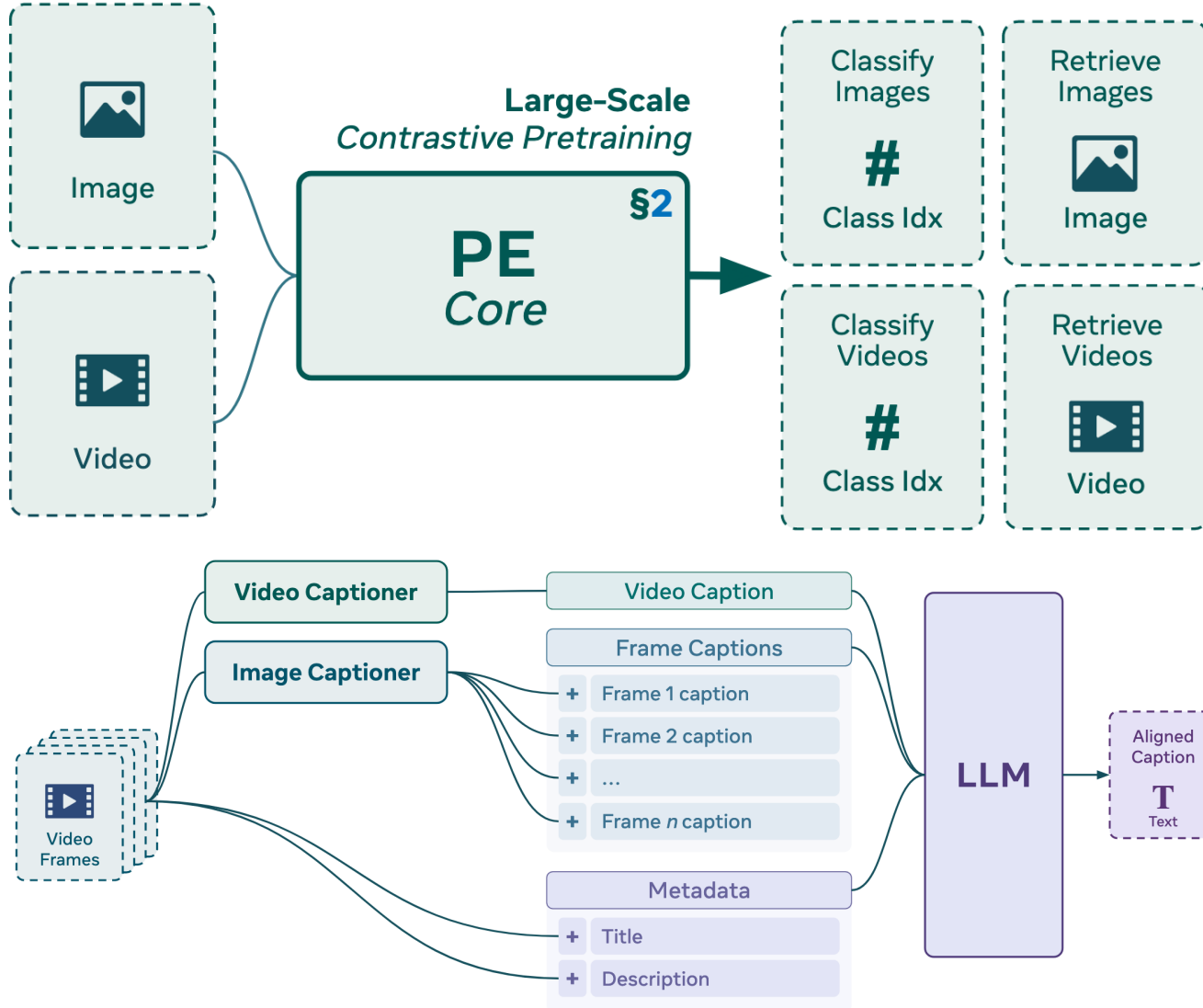


Figure 2. IMAGEBIND overview. Different modalities occur naturally aligned in different data sources, for instance images+text and video+audio in web data, depth or thermal information with images, IMU data in videos captured with egocentric cameras, *etc.* IMAGE-BIND links all these modalities in a common embedding space, enabling new emergent alignments and capabilities.

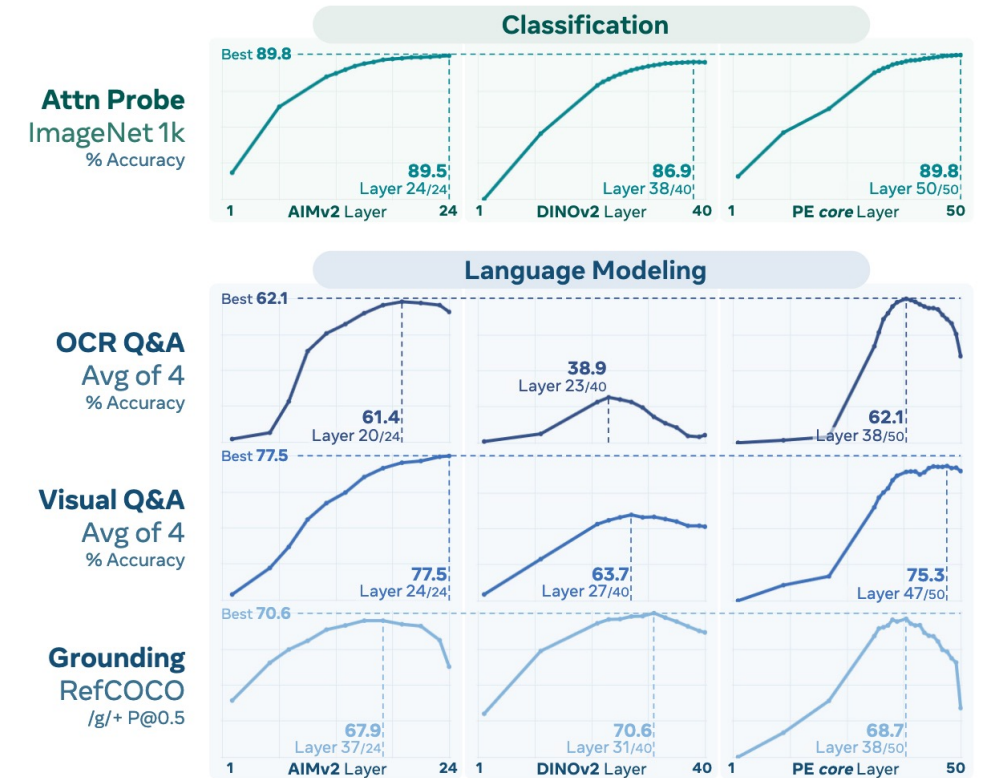
- Binds a given modality to image modality
- InfoNCE loss.
$$L_{\mathcal{I}, \mathcal{M}} = -\log \frac{\exp(\mathbf{q}_i^T \mathbf{k}_i / \tau)}{\exp(\mathbf{q}_i^T \mathbf{k}_i / \tau) + \sum_{j \neq i} \exp(\mathbf{q}_i^T \mathbf{k}_j / \tau)}$$
- ViT-H 630M params; text encoders (302M params) from OpenCLIP (frozen)
- Same encoder for images+videos. Treat videos as multi-frame images.

- Datasets:
 - (video, audio) pairs from Audioset
 - (image, depth) pairs from SUN RGB-D
 - (image, thermal) pairs from LLVIP
 - (video, IMU) pairs from Ego4D
 - (image, text) pairs from large-scale web data.

Perception Encoders

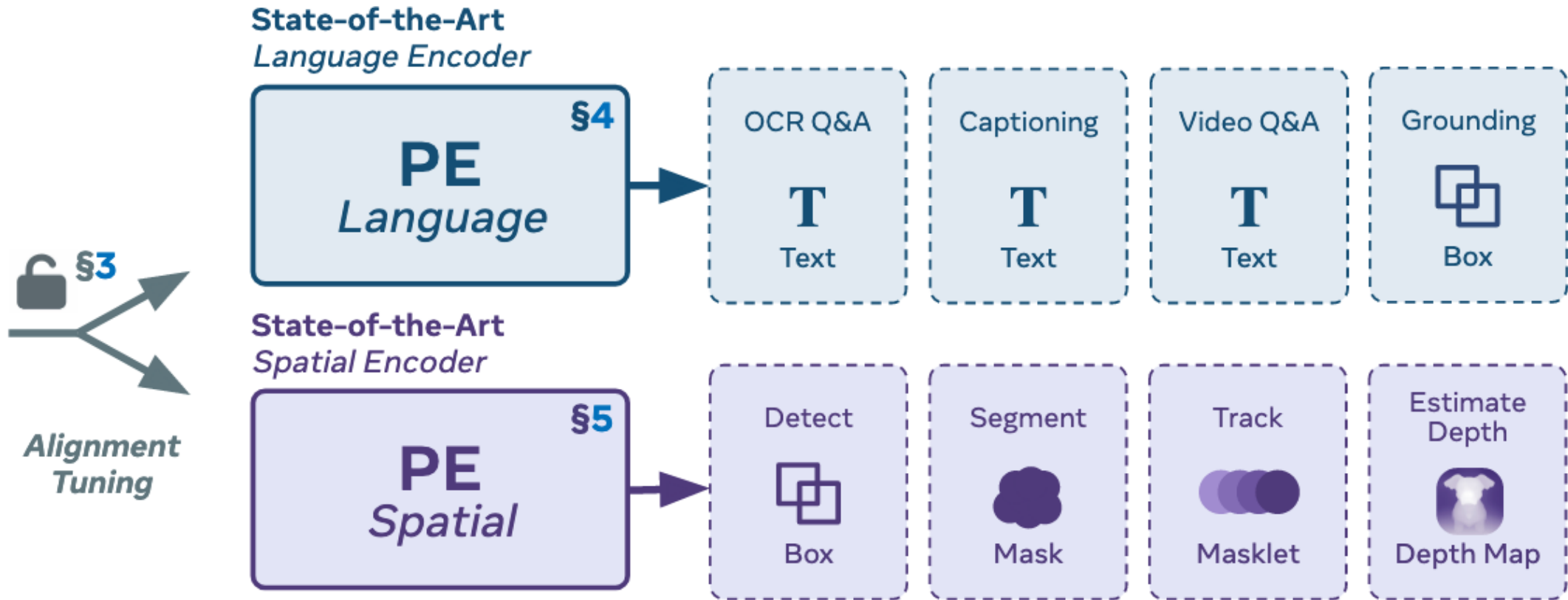


- First image-text (CLIP style + optimization)
- Then create synthetic captions for video and extend to video



Perception Encoders

Alignment tuning to obtain specific encoders which are good for different tasks (language and spatial)

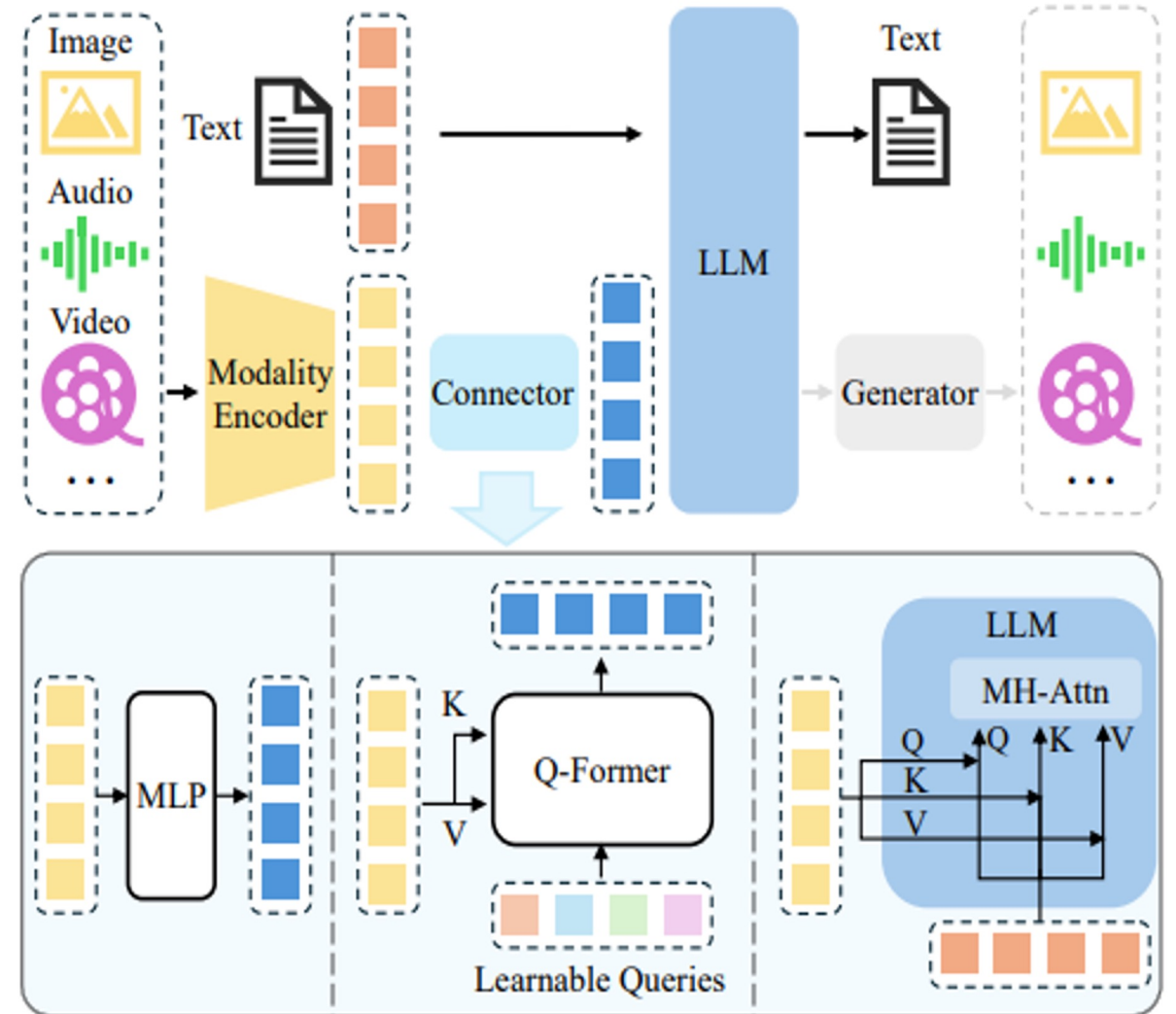


Multimodal Large Language Models

Architecture

A typical MLLM can be abstracted into three modules:

- a pre-trained modality encoder
- a pre-trained LLM
- a modality interface to connect them



Modality Encoder

The encoders compress raw information, such as images or audio, into a more compact and meaningful representation. Rather than training from scratch, a common approach is to use a pretrained encoder that has been aligned to other modalities.

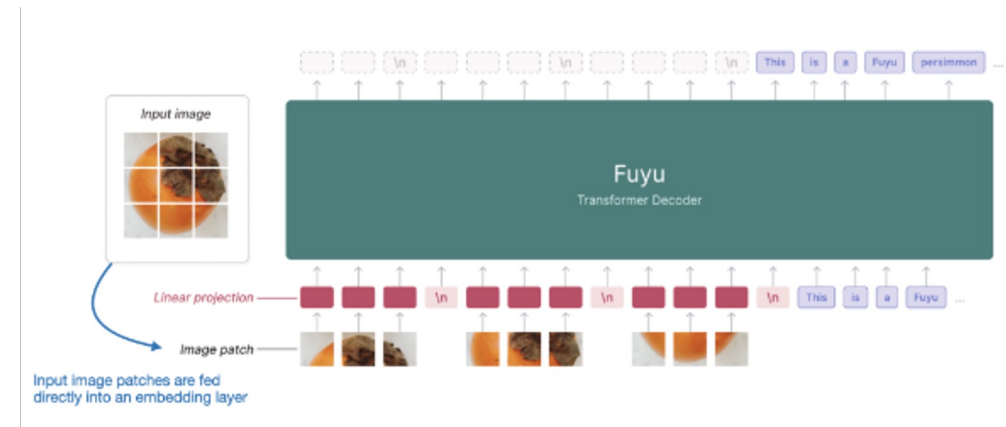
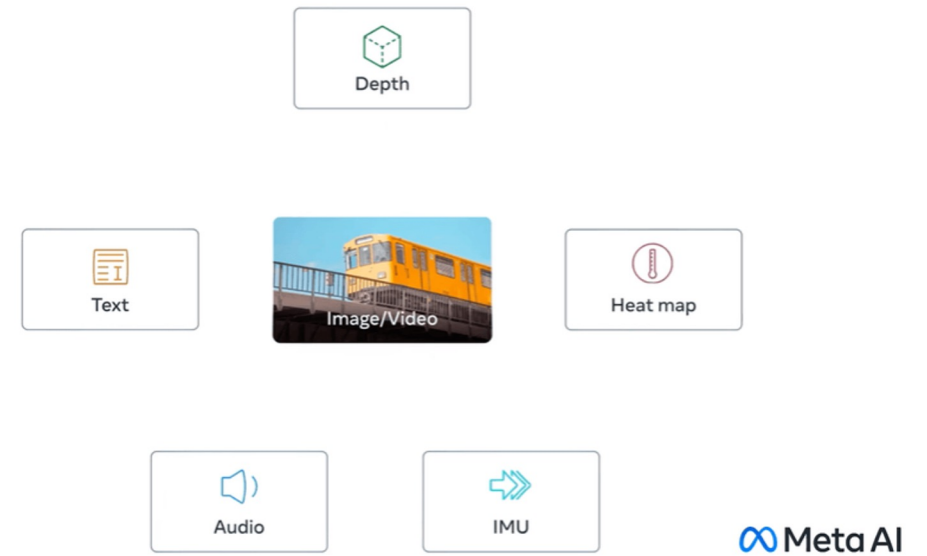
Commonly used image encoders are often CLIP-based ViT models.

Many works explored alternatives like EVA-CLIP, SigLIP, OpenCLIP-ViT or even OpenCLIP-ConvNext.

Notably Fuyu-8b (Bavishi et al., 2023) explored an encoder free architecture where the patches are simply projected by a linear layer to match text embedding dimension.

Ablation studies have demonstrated that scaling input resolution can achieve remarkable performance boost.

Of course we are not bound to the image world, Imagebind (Girdhar et al., 2023) is an encoder developed by Meta that supports images, text, audio, depth, IMU and thermal.



Connector

LLMs work at a token level and they can only perceive text, bridging the gap is necessary.

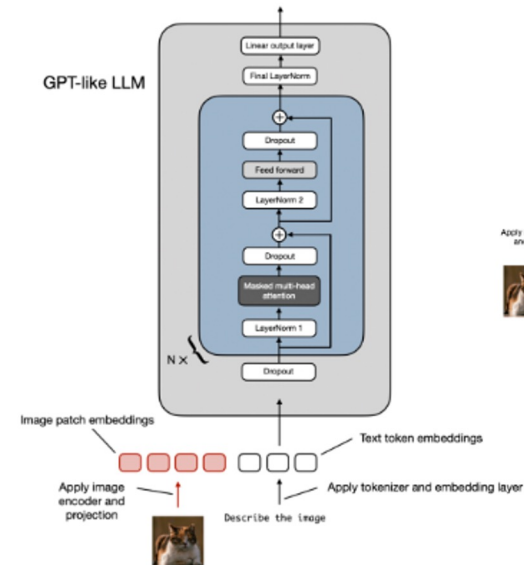
The connector is responsible to project the modality encoder features into the space that LLM can understand.

Two main methods exist:

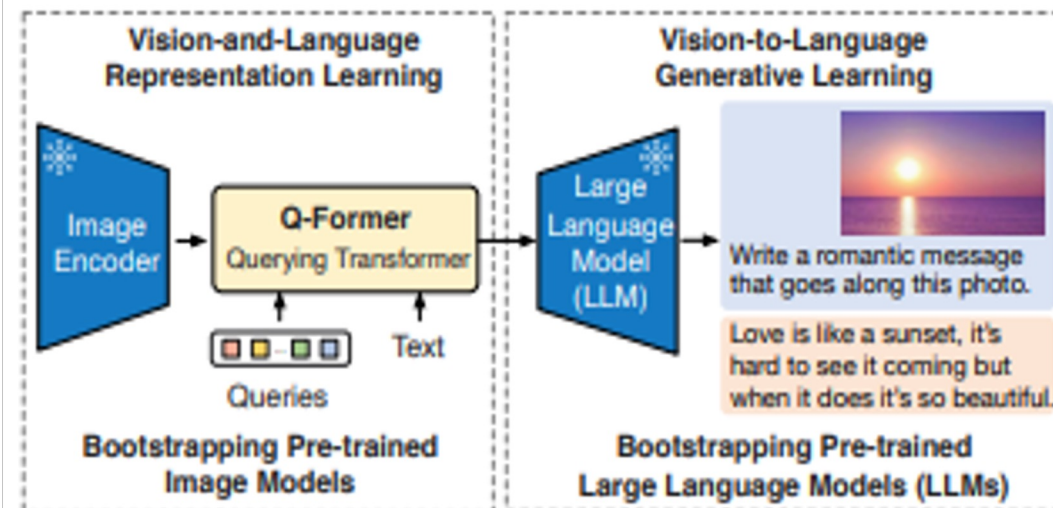
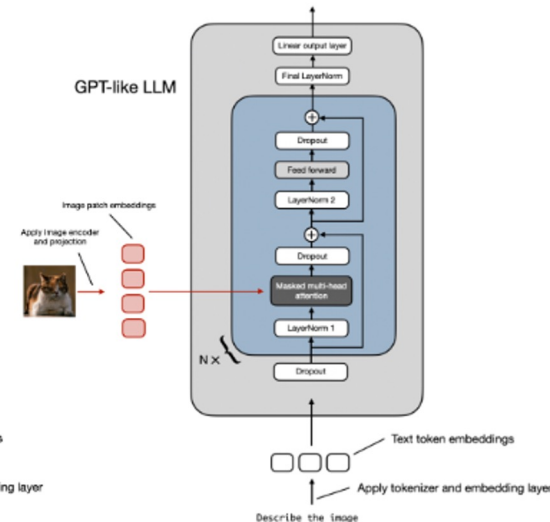
- **Unified embedding/Concatenation:** Just represents the image as a token series to be concatenated on the LLM input. Many papers have demonstrated that even a simple MLP is enough to do the job, Llava is an example (Liu et al., 2023). Ablation studies showed that when visual tokens are concatenated to the LLM input the connector choice is far less important than the number of visual tokens.
- **Cross Attention:** In this scheme the fusion happens inside the layers of the LLM, instead of adding new tokens to the input, text tokens attend through cross attention to the visual features,
- **Hybrid:** Blip-2 (Li et al., 2023) concatenates tokens to the input, but before it uses a Q-Former that does a form of cross-attention though, in this case the border blurs.

In terms of number of parameters the connector it's usually much smaller than the LLM or Encoder modules, in Qwen-VL (Bai et al., 2023) the Q-Former is about 0.08B (less than 1% of the whole parameters)

Method A: Unified Embedding Decoder Architecture



Method B: Cross-Modality Attention Architecture



Pre-training



- Almost every MLLM starts with pretrained components (encoder and LLM).
- Training an MLLM end-to-end is too expensive!
- The recipe is the same as for vanilla LLM: **Pre-training, Instruction tuning, Post-training.**
- Pre-training aims to align different modalities and learn multimodal world knowledge
- A common approach for pre-training is to **keep pre-trained modules (e.g. visual encoders and LLMs) frozen** and train a **learnable interface**, this is done to avoid losing any pre-trained knowledge.
- Given an image, the model is trained to predict autoregressively the caption of the image, following a standard cross-entropy loss.

Input: <image>
Response: {caption}

Instruction-tuning



- Instruction tuning aims to teach models to **better understand the instructions from users** and fulfill the demanded tasks.
- A multimodal instruction sample often includes an optional instruction and an input-output pair.
- The instruction is typically a natural language sentence describing the task, such as, “Describe the image in detail.” The input can be an image-text pair like the VQA task or only an image like the image caption task.
- Data adaptation is the key to collect data for this training stage. VQA and caption datasets are very useful but concise!
- With the release of GPT4V, many works have adopted GPT-4V to generate data of higher quality.
- Quality is no less important than quantity, many works demonstrated that tuning on smaller but high quality data leads to better performance than larger but noisy image-text pairs.

Below is an instruction that describes a task. Write a response that appropriately completes the request

Instruction: `<instruction>`

Input: `{<image>, <text>}`

Response: `<output>`

Post-training

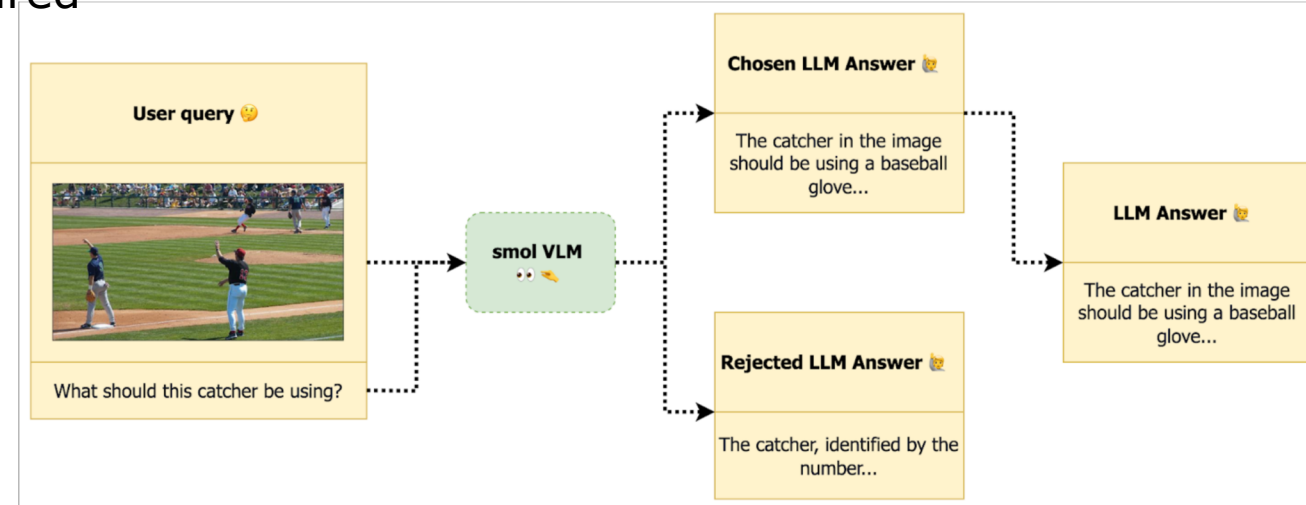


RLHF (Reinforcement Learning from Human Feedback):

- Collects human preference data
- Trains a reward model to estimate response quality
- Uses PPO to optimize the model based on estimated rewards

DPO (Direct Preference Optimization):

- Trains directly on preferred vs. rejected response pairs
- Loss based on the log-ratio of model probabilities
- No reward model or separate RL phase required



What to train/fine-tune and when

	Training Stages			ApolloBench					Overall
				OCR	Spatial	Egocentric	Perception	Reasoning	
1 stage		-	-	42.0	46.5	54.9	50.0	49.5	48.7
		-	-	28.8	29.2	18.8	35.5	22.6	30.8
2 stage			-	52.2	54.5	55.9	60.3	58.4	56.3
			-	51.6	54.5	58.0	62.1	60.2	<u>57.8</u>
			-	42.2	48.9	61.7	43.7	52.2	48.1
3 stage				53.0	52.5	64.9	59.8	65.9	59.2
				44.2	37.5	43.9	56.6	38.5	44.2

Classic multimodal tasks

Captioning: the model is asked to generate a descriptive sentence for an image.

VQA (Visual Question Answering): the flagship benchmark for vision-language tasks.

Visual Reasoning: Datasets like NLVR2 (Natural Language for Visual Reasoning) provide two images and a sentence, and ask whether the sentence is true with respect to those images.

Grounding: the task of linking language to specific regions or elements in an image or video, such as identifying the location of an object mentioned in a sentence.



Input Prompt

the left eye of emoji

Question: How many cows are here? Answer:

Question: What does the sign say? Answer:

Completion

(1)

(2)

(3)

FALSE

- Two penguins stand near each other in the picture on the left.
- There are only two penguins in at least one of the images.
- An image features two penguins standing close together.
- There are two penguins in the left image.
- An image contains just two penguins.
- There are four penguins

Quantization



Reduces weight precision (float32 → int8/int4)

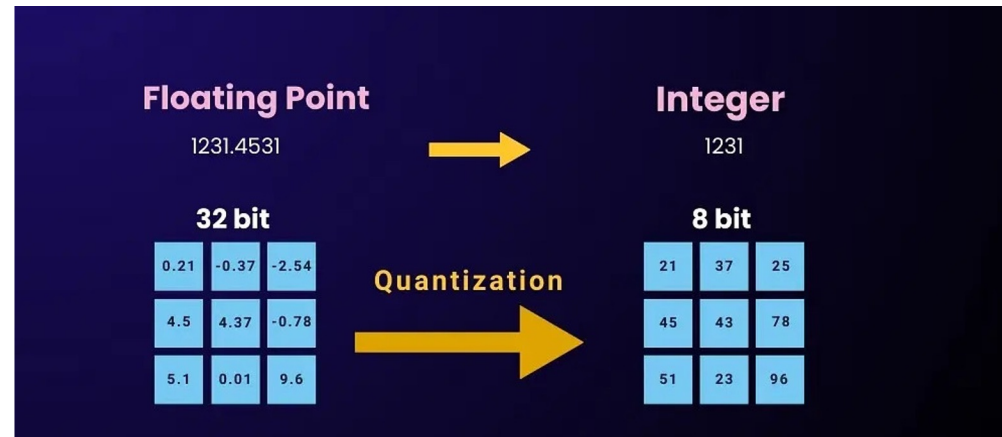
Enables inference on edge devices and adapter fine-tuning on quantized models.

bitsandbytes, zero-shot (no calibration), supports multimodal models (8-bit and 4-bit).

GPTQ (Generative Perturbation Quantization) is an offline method that leverages approximate second order information. It can quantize models up to 2-bit!

AWQ (Activation-Aware Weight Quantization) it is a newer way to quantize and leverages the idea that not all weights are born the same, it analyzes the activations distribution to protect the most critical ones.

GGUF is not thought to be directly used through the transformer library but to be used through software like Llama.cpp or LMStudio, you can recognize it because it uses suffixes like Q4_0 or Q5_K.



Main Video MLLMs (March 2026)



Open Source

Model Name	Base LLM	Parameters	Key Benchmark (VideoMME)	Key Feature
Qwen3-VL	Qwen3	8B, 32B, 235B	84.2% (72B)	Native 256K interleaved context
InternVL 3.5	InternLM 3	8B, 78B	74.0%	Strongest open-source reasoning
Video-SALMONN 2+	Llama 4	72B	81.6%	Best audio-visual temporal alignment
LLaVA-OneVision 2	Llama 4 / Qwen3	7B, 72B	71.2%	Training-free token compression
MiniCPM-o 2.6	-	8B	67.9%	Optimized for mobile/edge devices

Proprietary

Model Name	Key Benchmark (VideoMME Score)
GPT-5.2 Thinking	90.5%
Gemini 3.1 Pro	87.6%
Claude Opus 4.6	82.1%

The End